DELAB-IIITM WMT26: Fine-tuning NLLB-200 and IndicTrans2 for English–Low-Resource Indic Translation

Abstract

This paper describes DELAB-IIITM's submission system for the WMT26 Machine Translation shared task. We participated in the English-to-low-resource Indic translation direction, covering four language pairs: English to Assamese (Bengali), English to Manipuri (Bengali), English to Manipuri (Meitei Mayek) and English to Bodo (Devanagari). Our fine-tuning process leverages two pre-trained multilingual models: NLLB-200-Distilled-600M (Meta AI) and IndicTrans2-1.1B (AI4Bharat). For NLLB-200, we performed full fine-tuning on English-to-Assamese and English-to-Manipuri directions. For IndicTrans2, we employed Low-Rank Adaptation (LoRA) on English-to-Manipuri and English-to-Bodo directions, reducing trainable parameters from 1.1B to approximately 5.9M, enabling training on a single 24GB GPU. All models are fine-tuned exclusively on the WMT26-provided training data.

Reproducibility Code

Our complete codebase, including data preprocessing, fine-tuning scripts and LoRA configurations is publicly available at: https://github.com/Dingku99/DELAB-WMT26

Approach Details

Models Used

We used two pre-trained multilingual models:

- **NLLB-200-Distilled-600M**: A 600M parameter model covering 200+ languages, fine-tuned on English-to-Assamese(Bengali) and English-to-Manipuri (Bengali)
- **IndicTrans2-1.1B**: A 1.1B parameter model specifically designed for 22 Indic languages, fine-tuned on English-to-Manipuri (Meitei Mayek) and English-to-Bodo (Devanagari).

Fine-tuning Strategies

- **Full Fine-tuning (NLLB-200)**: All parameters updated with learning rate 2e-5, batch size 4, 3 epochs, fp16 mixed precision
- **LoRA Adaptation (IndicTrans2)**: Rank 16, alpha 32, dropout 0.1, target modules (q_proj, k_proj), learning rate 2e-4, effective batch size 8, gradient checkpointing enabled, float32 precision

Training Data

All models were trained exclusively on the WMT26-provided training data without any external augmentation. The training data statistics are as follows:

- English-to-Assamese (Bengali): 45,231 training sentences
- English-to-Manipuri (Bengali): 28,947 training sentences
- English-to-Manipuri (Meitei Mayek): 28,947 training sentences
- English-to-Bodo (Devanagari): 32,184 training sentences

Training Infrastructure

All experiments were conducted on a single NVIDIA RTX 3090 GPU with 24GB VRAM. Training time was approximately 4-6 hours for NLLB-200 and 8-10 hours for IndicTrans2 with LoRA for each language pairs.

Submitted Systems

We submitted two systems for each language pair:

- **Primary System:** Our baseline submission using the pre-trained model without any fine-tuning (i.e., the original IndicTrans2 and NLLB-200 models without adaptation).
- **Contrastive System:** Our fine-tuned models, which were adapted to the target language using the configurations described in Section 3. These include:
  - For NLLB: Full fine-tuning with standard hyperparameters (3 epochs, fixed learning rate)
  - For IndicTrans2: LoRA-based adaptation with rank $r = 16$, alpha $\alpha = 32$, and 3 epochs of training

The contrastive systems were designed to evaluate the effectiveness of fine-tuning, comparing the performance of the adapted models against their base counterparts. This allows us to quantify the improvements achieved through both full fine-tuning (NLLB) and parameter-efficient fine-tuning (IndicTrans2 with LoRA).